# VEXARDRIVE TECHNOLOGIES

## Data Scientist Intern Assessment

# Technical Analysis Report

Driver Behavior Risk & Vehicle Health Analytics

| | | | |
|---|---|---|---|
| **30**<br>DRIVERS | **30**<br>VEHICLES | **450**<br>TRIPS | **~12,987**<br>TELEMETRY RECORDS |

**Prepared for:** VexarDrive Technologies

**Assessment:** Data Scientist Intern

# Executive Summary

## Objective

This project analyzes a week of telematics data (GPS + IMU) to produce a data-driven **Driver Behaviour Risk Score** and a **Vehicle Health Score**.

## Analytical Approach

The analysis identifies rapid deceleration, rapid acceleration, high rotational movement, and high-speed exposure using empirical thresholds derived from Exploratory Data Analysis (EDA). Because official operational bounds and speed limits are unavailable, the methodology uses dataset-relative thresholds and exposure-normalized rates.

## Key Outputs

| | | | |
|---|---|---|---|
| **30 Drivers** | **30 Vehicles** | **450 Trips** | **~12,987 Telemetry Records** |

The final dashboards distinguish relative behavioural risk classifications and identify vehicles exhibiting persistent abnormal sensor signatures that may warrant maintenance inspection.

The generated scores are relative analytical indicators derived from telemetry. They are not externally certified safety ratings or confirmed mechanical-failure diagnoses.

# Data & Analytical Architecture

## Table Relationships

| | | |
|---|---|---|
| Telemetry | Trip_ID | Trips |
| Trips | Driver_ID | Drivers |
| Trips | Vehicle_ID | Vehicles |

*Trips.Vehicle_ID represents the actual vehicle associated with each trip.*

## Dataset Quality

After correcting workbook header formatting and validating data types and relationships, no missing values or corrupt records affecting the analytical metrics were identified.

# Feature Engineering

The raw data lacks explicit phone orientation calibration and official speed-limit context. Three core engineered features are therefore used.

## Acceleration Magnitude

```
Accel_Magnitude = sqrt(Accel_X_g^2 + Accel_Y_g^2 + Accel_Z_g^2)
```

**Purpose:** Provides an orientation-independent measure of combined accelerometer magnitude. **Interpretation:** Because gravity and motion are both present and device orientation is unknown, this is treated as a sensor-motion indicator rather than a direct mechanical-vibration or physical-force measurement.

## Gyroscope Magnitude

```
Gyro_Magnitude = sqrt(Gyro_X_dps^2 + Gyro_Y_dps^2 + Gyro_Z_dps^2)
```

**Purpose:** Acts as an orientation-independent rotational-movement indicator used as a cornering or swerving proxy.

## Minute Speed Change

```
Speed_Change_per_min = Speed_kmph(t) - Speed_kmph(t-1)
```

**Purpose:** Used to proxy acceleration and deceleration events. **Limitation:** Minute-to-minute speed changes do not represent instantaneous physical acceleration.

> **Important Interpretation Note:** Sensor-derived values are analytical proxies and should not be interpreted as direct mechanical measurements.

# Driver Behavior Methodology

## 1. Exposure-Normalized Metrics

Trip features are aggregated per driver and converted to rates per 100 telemetry minutes to reduce bias caused by different trip durations and observation volume.

## 2. Dataset-Derived Thresholds

- Rapid deceleration threshold: **-25.50 km/h/min**
- Rapid acceleration threshold: **+25.40 km/h/min**
- Rotational/cornering threshold: **7.58 dps**

These thresholds are empirical and dataset-relative, not industry-standard safety limits.

## 3. High-Speed Exposure

- **Source:** Speed_kmph
- **Threshold:** strictly greater than 40 km/h
- **Event:** one telemetry minute above 40 km/h
- **Metric:** high-speed minutes / total driving minutes

Because official speed limits are unavailable, the metric is explicitly labelled **High-Speed Exposure**, not legal speeding.

## 4. P95-Relative Normalization

```
Score_Raw = 100 * (Rate / P95) | Score = min(Score_Raw, 100)
```

Each component is scaled relative to its fleet 95th-percentile reference and capped at 100, limiting the influence of extreme observations while preserving relative differences.

## 5. Weighted Risk Score

| | |
|---|---|
| Braking / Deceleration | 30% |
| Acceleration | 30% |
| Corner / Rotation | 20% |
| Speed Behaviour | 20% |
| Total | 100% |

```
RiskScore = 0.30B + 0.30A + 0.20C + 0.20S
```

# Driver Score & Traceability

## Example: Driver D01

| Component | Subscore | Weight | Contribution |
|---|---|---|---|
| Braking | 27.427 | 30% | 8.228 |
| Acceleration | 22.066 | 30% | 6.620 |
| Corner | 82.961 | 20% | 16.592 |
| Speed | 34.915 | 20% | 6.983 |

```
Risk Score = (27.427 * 0.30) + (22.066 * 0.30) + (82.961 * 0.20) + (34.915 * 0.20)
```

**Final Score: 38.42** **Classification: Moderate**

The final score uses underlying unrounded values; displayed component values are rounded for readability.

## Interpretability

The score is decomposable into four observable behavioural dimensions. This allows the evaluator to trace the final value rather than treating the score as a black-box prediction.

# Vehicle Health Methodology

The vehicle analysis identifies persistent abnormal acceleration-magnitude signatures that may warrant inspection. It does not diagnose a specific mechanical fault.

## Method

- Compute mean acceleration magnitude for each trip.
- Flag a trip when its mean exceeds the fleet **90th percentile = 1.046 g**.
- Calculate **Persistence = anomalous trips / total trips × 100** per vehicle.
- Use the fleet **75th percentile of persistence = 13.33%** as the persistence threshold.
- Require anomaly evidence across more than one driver before assigning Maintenance Recommended.

## Cross-Driver Rule

The cross-driver condition reduces the likelihood of attributing driver-specific behaviour to the vehicle.

```
IF Persistence > 13.33% AND Drivers_With_Anomaly > 1 THEN Maintenance Recommended ELSE IF
Persistence > 13.33% AND Drivers_With_Anomaly = 1 THEN Monitor
```

These thresholds are fleet-relative and dataset-derived; they are not industry-standard mechanical maintenance thresholds.

# Vehicle Validation

## V02 / V19 Comparison

| Vehicle | Trips | Anomalous Trips | Persistence | Drivers | Classification |
|---|---|---|---|---|---|
| V02 | 16 | 8 | 50.00% | 2 | Maintenance Recommended |
| V19 | 16 | 7 | 43.75% | 1 | Monitor |

V19 exceeds the persistence threshold, but its anomalous trips come from only one driver, so it is classified as **Monitor**. V02 satisfies both persistence and cross-driver criteria and is therefore assigned **Maintenance Recommended** priority for further inspection.

# Dashboard Insights

The interactive dashboards are delivered separately through the hosted dashboard link and exported dashboard images. This report summarizes their analytical interpretation rather than duplicating the full dashboard visuals.

## Driver Behaviour Dashboard

The dashboard ranks drivers by relative risk score, shows the fleet risk distribution, and provides component-level drill-down for braking, acceleration, cornering, and high-speed exposure.

**Key analytical question**
Which drivers exhibit comparatively higher-risk behavioural patterns, and which behavioural components drive their scores?

## Vehicle Health Status Dashboard

The vehicle dashboard compares anomaly persistence and cross-driver evidence, helping separate vehicles requiring inspection from cases where abnormal telemetry is concentrated under a single driver.

**Key analytical question**
Which vehicles show persistent abnormal sensor signatures across multiple drivers and therefore deserve inspection priority?

Dashboard evidence is traceable to the generated CSV metrics and the documented scoring rules. No dashboard status is presented as a confirmed safety violation or mechanical diagnosis.

# Assumptions & Limitations

## Assumptions

- Accelerometer axis orientation is unknown.
- 3D magnitude is used as an orientation-independent sensor-motion indicator.
- Minute-level speed changes are behavioural proxies.
- Official speed limits are unavailable.
- Thresholds are empirical and dataset-relative.
- Cross-driver persistence reduces, but does not eliminate, driver confounding.
- Risk and health scores are relative analytical indicators.

## Limitations

- **Temporal resolution:** one-minute telemetry misses short-duration events.
- **Ground truth:** no risky-driver or mechanical-failure labels are available.
- **Context:** no road, weather, traffic, gradient, or route context.
- **Sensor orientation:** no phone mounting calibration.
- **Diagnosis:** abnormal signatures require physical inspection.

## Additional Applications

| Application | Current Data | Additional Data Needed |
|---|---|---|
| Driver Coaching | Yes | — |
| Fleet Safety Monitoring | Yes | — |
| Maintenance Prioritization | Yes | Maintenance outcomes for validation |
| Predictive Maintenance | Partial | Failure / maintenance labels |
| Insurance Risk Analysis | Potential | Claims / accident outcomes |

## Conclusion

The assessment demonstrates an explainable, exposure-aware approach to extracting driver-behaviour and vehicle-maintenance signals from GPS and IMU telemetry while explicitly accounting for data limitations and driver/vehicle confounding. The resulting framework provides defensible prioritization for fleet safety review and maintenance inspection.